

Singapore Birth and Fertility Time Series Analysis

MATH X313 — Time Series Analysis

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1 Research Question

We use the Total Fertility Rate and the Total Live Births data from the Singapore Department of Statistics website, hereafter abbreviated as TFR and TLB respectively. Singapore's TFR has fallen from 5.76 children per woman in 1960 to 0.87 in 2025 — the lowest on record and far below the replacement level of 2.1. Over the same period, TLB declined from 61,775 to 30,004. This sustained decline is not unique to Singapore: Shair et al. (2024) document a similarly dramatic fall in Malaysia's TFR from an average of 6.4 infants per woman in 1960 to 1.8 in 2020, and project Chinese and Indian Malaysian fertility rates falling below 1.0 by 2040.

Understanding which population subgroups drive TFR is essential context for interpreting these trends. Chen et al. (2018) conduct an elasticity analysis of Singapore's TFR and find that 20–29-year-old single women and married childless women aged 30–34 are the most influential groups for changing TFR, yet Singapore's pro-natalist policies have historically favoured third and higher-order births — a policy mismatch that limits their effectiveness. The inability of targeted interventions to reverse the decline motivates a data-driven time series approach to quantify the trajectory statistically and produce reliable near-term projections.

Research question: How have Singapore’s annual TFR and TLB evolved since 1960, and which course-based time series models provide the most reliable short-term forecasts of each series after residual diagnostic checking?

The question is suited to annual time series methods because both series are measured once per year over a long observation window (66 years), exhibit clear trend and autocorrelation structure, and do not contain a seasonal component that would require higher-frequency data.

2 Data

Source and Variables

The data are drawn from the Singapore Department of Statistics (SingStat) births and fertility register, supplied as the file `data/raw/BirthsAndFertilityRatesAnnual.csv`. The file is in wide format (one row per demographic series, one column per year); the preparation script `scripts/01_prepare_data.R` reshapes it into a long-format tibble indexed by measure and year.

Two series are retained for analysis:

- **Total Fertility Rate (TFR):** the average number of children a woman would bear over her lifetime at the age-specific fertility rates observed in a given year. Measured in children per woman.
- **Total Live Births:** the count of live births registered in Singapore in each calendar year.

Table 1: Analysis series summary

Measure	First year	Last year	N	First observation	Latest observation
Total fertility rate	1960	2025	66	5.76	0.87
Total live births	1960	2025	66	61775	30004

Data Validation

An automated validation script confirmed: both series present, no missing observations, no duplicate measure-year keys, years consecutive from 1960 to 2025, and all values strictly positive. No data quality corrections were required.

Table 2: Data validation checks

check	passed	detail
<code>expected_measures_present</code>	TRUE	Total fertility rate; Total live births
<code>no_missing_values</code>	TRUE	0 missing observations
<code>no_duplicate_measure_year_keys</code>	TRUE	0 duplicate keys
<code>annual_years_are_consecutive</code>	TRUE	0 series with non-consecutive years
<code>positive_observations</code>	TRUE	0 non-positive observations
<code>minimum_history_length</code>	TRUE	minimum observations per measure: 66

The two series are highly correlated (Pearson $r = 0.847$), confirming that they move together over the long run while remaining analytically distinct. All subsequent analyses treat TFR and Total Live Births as separate univariate time series.

3 Exploratory Analysis

Level Series

Annual Singapore fertility and birth series

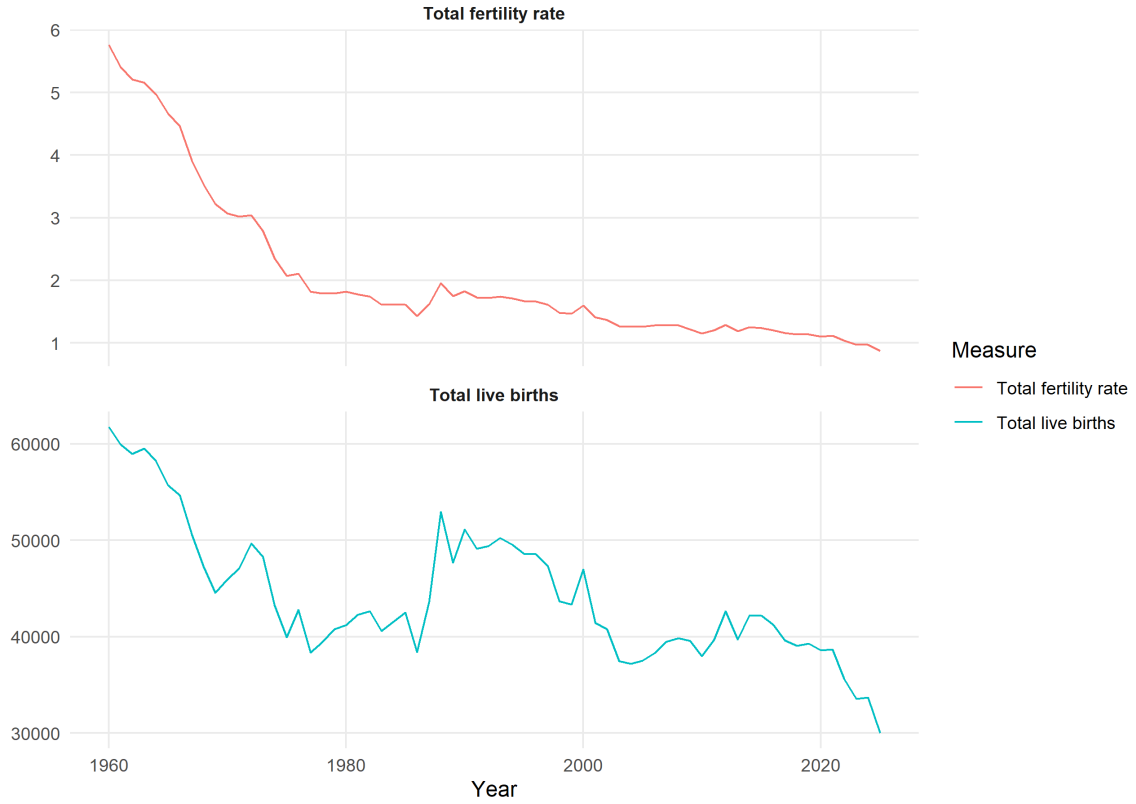


Figure 1: Annual TFR and Total Live Births, Singapore 1960–2025.

Both series decline persistently from 1960 to 2025 (@fig-overview). TFR falls steeply through the 1960s and early 1970s as Singapore’s post-independence demographic transition accelerated, then levels off near replacement level in the 1980s and 1990s before resuming a gradual decline. Total Live Births mirrors this broad pattern, with a notable baby-boom rebound in 1988 (the largest single-year increase of 9,341 births) followed by continued decline to the record low of 30,004 in 2025. Neither series is stationary: the mean level shifts persistently over time.

Table 3: Descriptive statistics by sub-period

Series	Period	N	Mean	SD	CV
Total fertility rate	early	22	3.35182	1.37165	0.409225
Total fertility rate	middle	22	1.61864	0.16149	0.0997691
Total fertility rate	recent	22	1.16455	0.115706	0.0993572
Total live births	early	22	48646.2	7633.62	0.156921
Total live births	middle	22	45344.3	4396.3	0.0969538
Total live births	recent	22	38435.9	3027.29	0.0787622

The coefficient of variation ($CV = SD/\text{mean}$) for TFR declines sharply across the three 22-year sub-periods (early 1960–1981: $CV = 0.41$; middle 1982–2003: $CV = 0.10$; recent 2004–2025: $CV = 0.10$), indicating that the large early variation is not proportional to the mean level. For

Total Live Births the CV is more stable ($0.16 \rightarrow 0.10 \rightarrow 0.08$), consistent with a series where variance scales more gently with the level.

Annual First Differences

Year-to-year changes

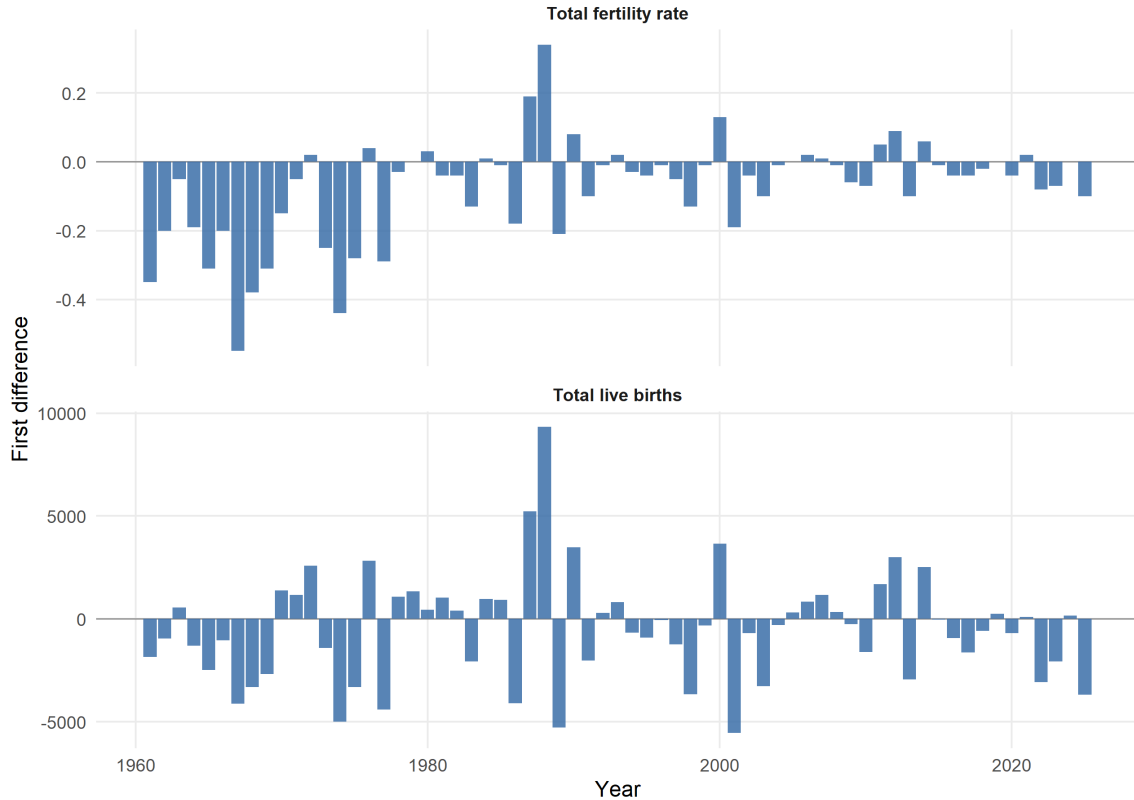


Figure 2: Year-on-year first differences of TFR (top) and Total Live Births (bottom).

@fig-diffs shows the year-on-year changes. TFR has a mean annual change of -0.075 (SD 0.146), with the largest single-year decrease of -0.55 in 1967 and the largest increase of $+0.34$ in 1988. Total Live Births averages -489 per year (SD $2,595$). The 1988 baby-boom produced the largest single-year increase of $9,341$; the sharpest single-year decline of $-5,546$ occurred in 2001, following the end of the baby-boom cohort entering peak childbearing age.

The first differences appear considerably more stationary than the level series, suggesting that one order of differencing will be sufficient for modelling.

4 Transformation and Stationarity

Scale Selection

All observations are strictly positive, so logarithmic transformation is available for both series. @fig-scale overlays original and log-scaled (indexed) series. For TFR, the CV collapses from 0.41 in the early period to 0.10 in later periods, indicating non-proportional variance; the original scale is therefore preferred. For Total Live Births, the more stable CV ($0.16 \rightarrow 0.08$) means that log scale could also be used, but the original scale is retained throughout to maintain comparability between the two series. Log-scale modelling of live births is noted as an avenue for future model extensions.

Original-scale and log-scale standardised series

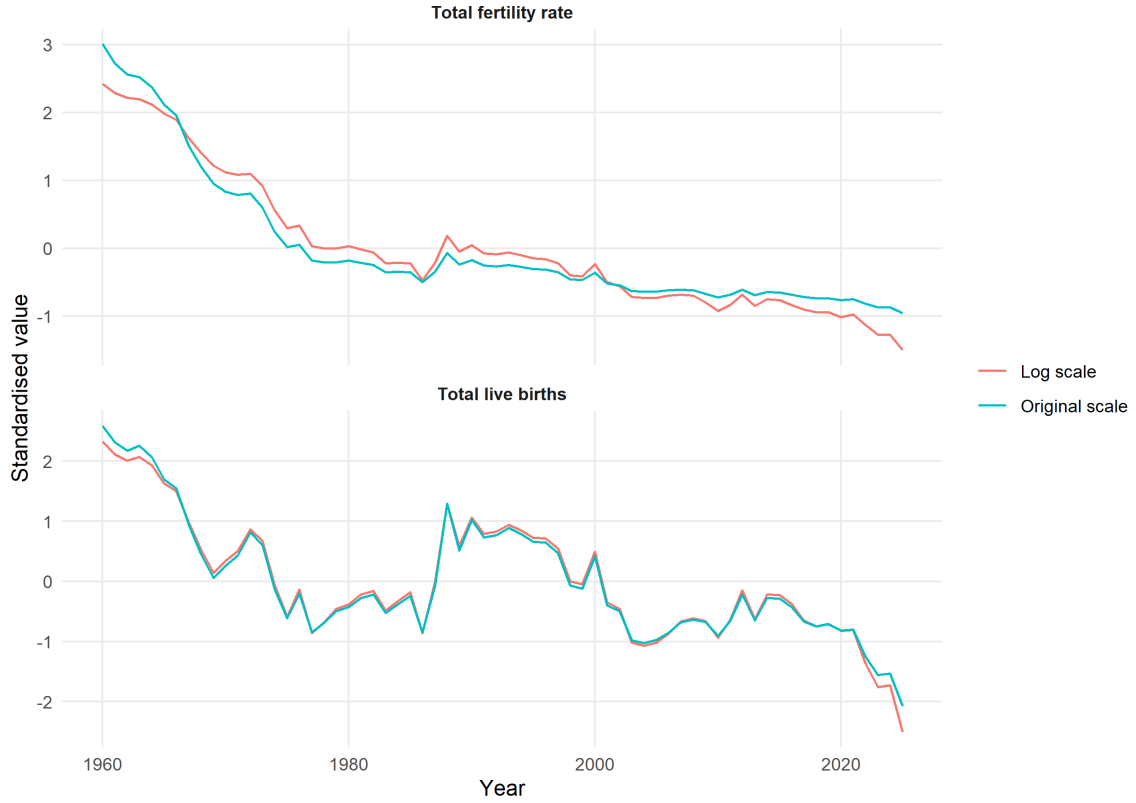


Figure 3: Original scale and log-scale standardised series, 1960 = 1.

Autocorrelation Evidence

Table 4: Sample ACF1 at the level and after successive differencing

Measure	level_acf1	first_difference_acf1	second_difference_acf1
Total fertility rate	0.914504	0.370068	-0.401155
Total live births	0.841296	-0.114754	-0.522989

Table 2 summarises the first-order autocorrelation coefficient (ACF1) at the level and after successive differencing. Both level series show very high ACF1 (0.915 for TFR, 0.841 for TLB), consistent with non-stationary or near-unit-root behaviour. Letting $\nabla y_t = y_t - y_{t-1}$ denote the first-difference operator, one order of differencing reduces ACF1 to 0.370 (∇ TFR) and -0.115 (∇ TLB)—values close enough to zero to support approximate stationarity. Second differencing overshoots, to -0.401 and -0.523 respectively), confirming over-differencing. These results justify using $\mathbf{d} = \mathbf{1}$ in all ARIMA candidates.

5 Candidate Models

Training and Holdout Design

The 66-observation sample is split into a **training window** (1960–2015, $n = 56$) and a **holdout window** (2016–2025, $n = 10$). All candidate models are estimated on the training data only. Holdout forecasts are compared to the observed values to evaluate out-of-sample accuracy.

Table 5: Training and holdout split

Measure	sample	first_year	last_year	observations
Total fertility rate	full_series	1960	2025	66
Total live births	full_series	1960	2025	66
Total fertility rate	training	1960	2015	56
Total live births	training	1960	2015	56
Total fertility rate	holdout	2016	2025	10
Total live births	holdout	2016	2025	10

The 10-year holdout represents approximately 15% of the full series, which is a reasonable proportion for an annual demographic series. A longer holdout would reduce the training sample and risk underfitting the long-run trend.

Model Set

Five candidate models are fitted for each series:

Table 6: Candidate model registry

model	model_class	specification	purpose
naive	benchmark	NAIVE(Observed)	Last-observation baseline for forecast comparison.
drift	benchmark	RW(Observed ~ drift())	Random-walk benchmark with average historical drift.
linear_trend	trend	TSLM(Observed ~ trend())	Simple deterministic trend benchmark.
auto_arima	ARIMA	ARIMA(Observed)	Course-based stochastic time series candidate.
ets	state-space	ETS(Observed)	Exponential smoothing state-space candidate.

The two **benchmark** models (Naïve and Drift) provide simple baselines. The Naïve model forecasts the last observed value; the Drift model adds the average historical annual change. Any adequately specified model should outperform at least one of these.

The **linear trend** model (TSLM) tests whether a deterministic regression on a time index is sufficient to explain the series. Its inclusion is valuable because it often achieves good holdout accuracy on monotonically declining series, but the residual structure must be checked carefully.

The **ARIMA** and **ETS** candidates are the two principal course-based stochastic model families. ARIMA (AutoRegressive Integrated Moving Average) models the differenced series as a linear combination of its own past values and past innovations; ETS (Error, Trend, Seasonal) uses an exponential smoothing state-space framework (Hyndman and Athanasopoulos, 2021). Orders and parameters are selected by minimising the corrected Akaike information criterion (AICc; Hurvich and Tsai, 1989), which penalises model complexity more aggressively than AIC in small samples.

Selected ARIMA Orders

Automatic order selection produced different structures for the two series:

TFR — ARIMA(0,1,1). The selected model applies one order of differencing ($d = 1$) and a single moving-average term ($q = 1$), with no autoregressive terms ($p = 0$). In the notation of Box et al. (2015), the fitted equation for $\nabla y_t = y_t - y_{t-1}$ is:

$$\nabla y_t = (1 - 0.799 B) \varepsilon_t, \quad \varepsilon_t \sim N(0, 0.028)$$

where B is the backshift operator ($By_t = y_{t-1}$) and ε_t is Gaussian white noise with estimated variance $\hat{\sigma}^2 = 0.028$. The MA(1) coefficient $\hat{\theta}_1 = -0.799$ is highly significant (SE = 0.075, $t \approx -10.7$, $p < 0.001$). The negative sign means a positive shock in period t is partially reversed in the following period — consistent with modest short-term mean-reversion in TFR around its declining long-run level. This ARIMA(0,1,1) is algebraically equivalent to simple exponential smoothing applied to the first-differenced series, with AICc = -54.80 .

Total Live Births — ARIMA(0,1,0). Automatic order selection chose a pure random walk — one order of differencing with no AR or MA terms:

$$\nabla y_t = \varepsilon_t, \quad \varepsilon_t \sim N(0, 6,811,530)$$

No additional parameters improved the AICc (= 1029.14). This specification is algebraically identical to the Naïve model (point forecast = last observed value), so the **Drift model** — which augments the random walk with an estimated constant annual change — provides a more informative alternative:

$$\nabla y_t = \hat{b} + \varepsilon_t, \quad \hat{b} = -356 \text{ births/year}$$

Table 7: Fitted model coefficients (training sample, 1960–2015)

Measure	.model	term	estimate	std.error	statistic	p.value
Total fertility rate	drift	b	-0.0821818	0.0212547	-3.86652	0.000298737
Total fertility rate	linear_trend	(Intercept)	4.05619	0.19113	21.2222	7.20365e-28
Total fertility rate	linear_trend	trend()	-0.0644655	0.00583348	-11.0509	1.76788e-15
Total fertility rate	auto_arima	ma1	-0.799258	0.0746695	-10.7039	5.82464e-15
Total fertility rate	ets	alpha	0.992637			
Total fertility rate	ets	beta	0.000100017			
Total fertility rate	ets	phi	0.900712			
Total fertility rate	ets	l[0]	6.19882			
Total fertility rate	ets	b[0]	-0.510731			
Total live births	drift	b	-356.182	370.585	-0.961134	0.340769
Total live births	linear_trend	(Intercept)	52541.6	1369.01	38.3794	7.11929e-41

Measure	.model	term	estimate	std.error	statistic	p.value
Total live births	linear_trend	trend()	-249.528	41.7835	-5.97194	1.89147e-07
Total live births	ets	alpha	0.889752			
Total live births	ets	l[0]	61569.9			

6 Model Diagnostics

Residual Time Plots

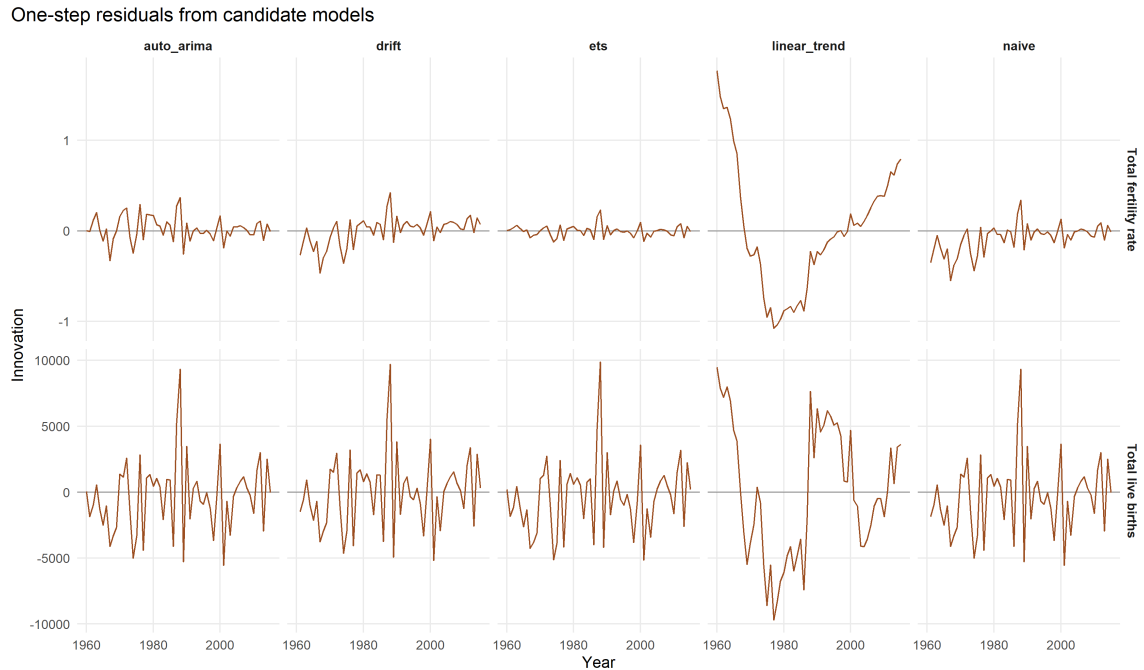


Figure 4: One-step training residuals by model (columns) and series (rows).

@fig-residuals shows the one-step innovations (residuals) on the training data. For TFR, the ARIMA(0,1,1) residuals scatter randomly around zero with no discernible pattern. The linear trend residuals show a pronounced U-shaped structure — positive in the early period, negative in the middle, then positive again — confirming that a purely deterministic linear trend fails to capture the changing rate of TFR decline. For Total Live Births, the ARIMA(0,1,0) and Naïve models produce identical residuals (as expected for equivalent specifications); the Drift model residuals are similarly consistent with white noise.

Residual ACF

@fig-resid-acf plots the residual autocorrelation functions to lag 14. For TFR, the ARIMA(0,1,1) residual ACF has no spikes outside the 95% confidence bands, providing strong visual evidence of adequate fit. The linear trend model leaves persistent positive autocorrelation at early lags — a clear sign of unmodelled dynamics. For Total Live Births, the ARIMA(0,1,0), Drift, ETS, and Naïve residual ACFs are all consistent with white noise; the linear trend again shows strong autocorrelation.

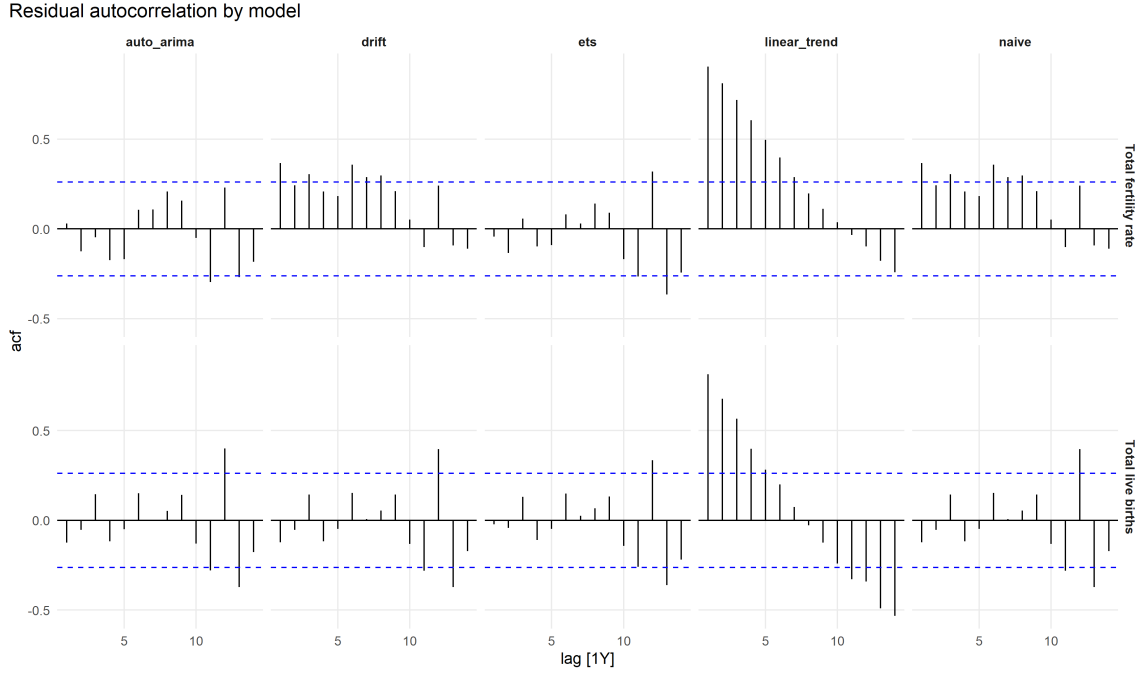


Figure 5: Residual autocorrelation functions (lags 1–14) for each candidate model.

Ljung-Box Tests

The Ljung–Box portmanteau test at lag 10 (Ljung and Box, 1978) provides a formal check of residual independence. A p-value above 0.05 indicates that the residuals do not show significant autocorrelation at the 5% level, which we take as evidence of adequate model fit.

Table 8: Ljung-Box residual test (lag = 10) for each candidate model

Series	Model	LB statistic	p-value
Total fertility rate	auto_arima	11.1487	0.34605
Total fertility rate	drift	44.5183	2.65594e-06
Total fertility rate	ets	6.88353	0.736393
Total fertility rate	linear_trend	177.473	0
Total fertility rate	naive	44.5183	2.65594e-06
Total live births	auto_arima	7.64524	0.663444
Total live births	drift	7.58541	0.669262
Total live births	ets	6.54603	0.767494
Total live births	linear_trend	109.597	0
Total live births	naive	7.58541	0.669262

For TFR, only ARIMA(0,1,1) ($p = 0.346$) and ETS ($p = 0.736$) pass the test. The Naïve and Drift models fail badly ($p < 0.001$), as does the linear trend model ($LB = 177.5$, $p < 0.001$). For Total Live Births, ARIMA(0,1,0), Drift, ETS, and Naïve all pass; the linear trend fails decisively ($LB = 109.6$, $p < 0.001$).

7 Forecast Evaluation

Holdout Accuracy

Table 9: Holdout forecast accuracy (2016–2025)

.model	Measure	RMSE	MAE	MAPE
auto_arima	Total fertility rate	0.178424	0.153068	15.2901
ets	Total fertility rate	0.190391	0.163577	16.334
naive	Total fertility rate	0.196138	0.169	16.8645
drift	Total fertility rate	0.317951	0.283	27.8023
linear_trend	Total fertility rate	0.984112	0.979432	92.9922
linear_trend	Total live births	2718.99	2321.32	6.63675
drift	Total live births	4097.4	3296.1	9.65776
ets	Total live births	6221.95	5233.22	15.2028
auto_arima	Total live births	6240.36	5255.1	15.2625
naive	Total live births	6240.36	5255.1	15.2625

For **TFR** (@tbl-holdout-accuracy), ARIMA(0,1,1) achieves the lowest holdout RMSE (0.178) and MAE (0.153) of all five candidates. ETS is the closest competitor (RMSE 0.190). The linear trend extrapolates a long-run trajectory that substantially overshoots the near-term observed values (MAPE 93%), making it the weakest performer despite its simple structure.

For **Total Live Births**, the linear trend achieves the lowest holdout RMSE (2,719) and MAE (2,321). However, its residuals carry severe autocorrelation ($LB = 109.6$, $p < 0.001$), which invalidates its prediction intervals — a model with correlated residuals will systematically misstate forecast uncertainty. Among models with adequate residual diagnostics, the **Drift** model achieves the best RMSE (4,097) and MAE (3,296), substantially better than ARIMA(0,1,0), ETS, and Naïve (all around RMSE 6,240).

Holdout Forecast Plot

@fig-holdout shows the training series, holdout observed values, and 80% prediction intervals for each candidate. For TFR, the ARIMA(0,1,1) band is well-centred on the observed holdout path. For Total Live Births, the linear trend tracks the holdout trajectory closely by extrapolating the long-run decline, but its prediction intervals are statistically invalid. The Drift model’s intervals are wider but properly calibrated.

Model Selection

TFR — ARIMA(0,1,1) selected. This model simultaneously achieves the best holdout RMSE and MAE and passes the Ljung-Box residual test ($p = 0.346$). No other candidate improves on both criteria. ETS is retained as a secondary reference in the forecast figures.

Total Live Births — Drift model selected. The linear trend’s superior raw accuracy is disqualified by a catastrophic residual diagnostic failure (LB statistic = 109.6, $p < 0.001$): its residuals carry substantial unmodelled autocorrelation that invalidates any stated prediction interval. Among the four diagnostically adequate models, Drift achieves the best holdout RMSE (4,097) and retains an economically interpretable structure — the estimated annual decline of 356 births per year directly reflects the observed long-run trend. Although the Drift coefficient for live births is not individually significant at the 5% level ($t = -0.96$, $p = 0.341$), the substantive direction of decline is supported by over six decades of data and is appropriate for a five-year projection horizon.

Holdout-period forecasts

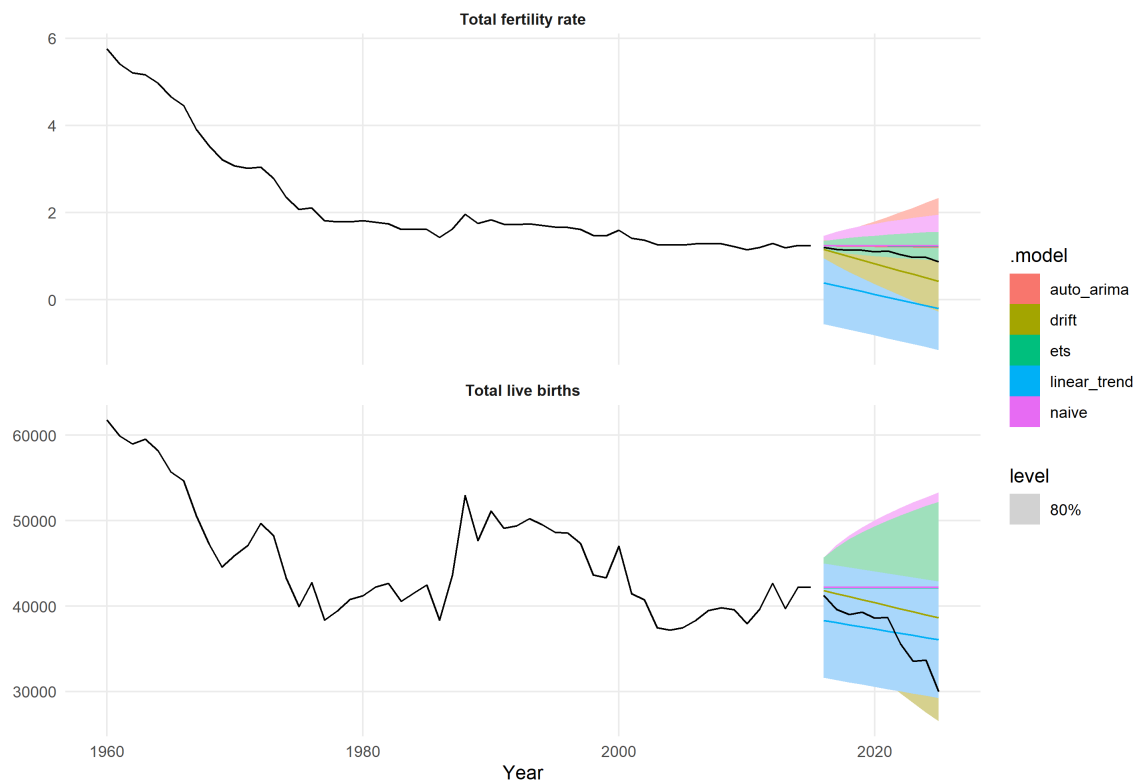


Figure 6: Holdout forecast comparison (2016–2025). Solid black: observed. Shaded: 80% prediction interval.

8 Final Forecasts

Both selected models are refitted to the full 66-observation series (1960–2025) before producing five-year-ahead forecasts. Using the complete sample rather than the training-only window reduces estimation uncertainty now that model selection is settled.

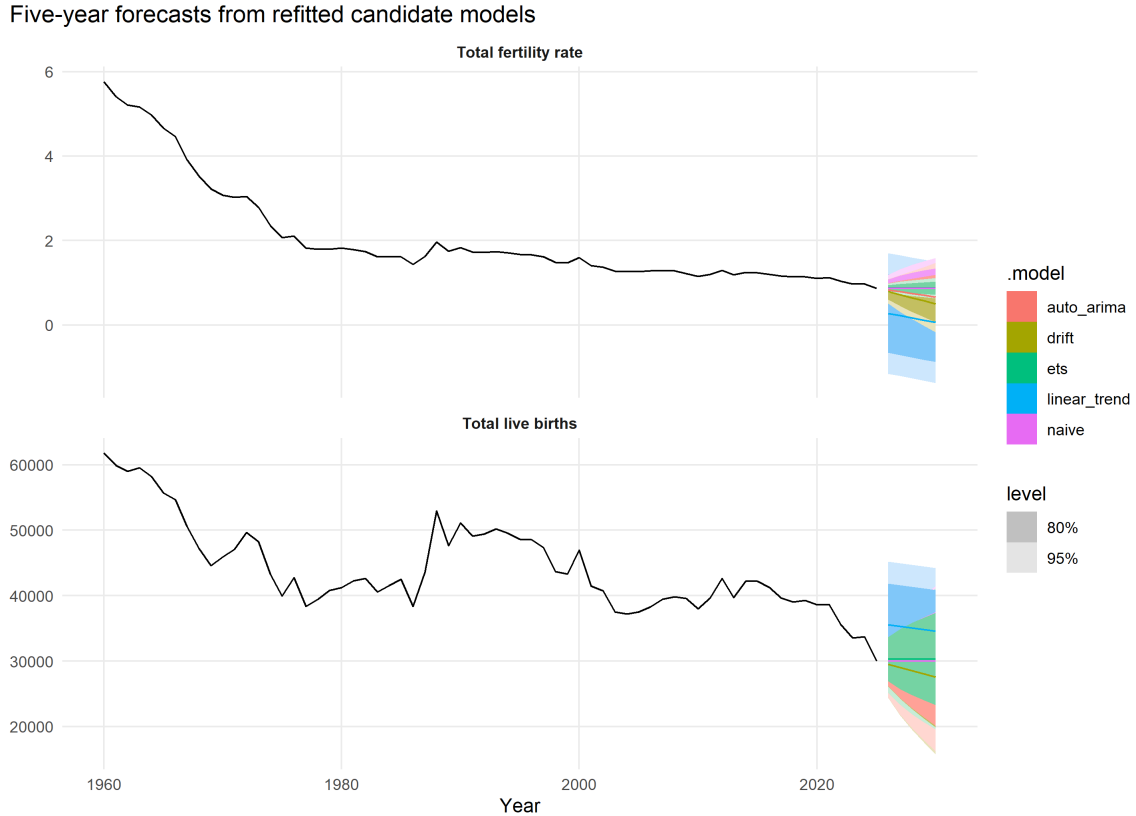


Figure 7: Five-year forecasts (2026–2030) from all candidates refitted to the full 1960–2025 series. Inner band: 80% PI; outer band: 95% PI.

Both selected models are refitted to the complete 66-observation series (1960–2025) before producing five-year-ahead forecasts (@fig-forecasts). The selected models — ARIMA(0,1,1) for TFR and Drift for Total Live Births — are the primary forecasting reference.

TFR. The ARIMA(0,1,1) model refitted to 1960–2025 projects a continued modest decline, with the point forecast for 2030 falling below 0.87. The 95% prediction interval widens considerably at five years because the random-walk component accumulates uncertainty each period. The lower bound of the 95% interval approaches zero by 2030, reflecting substantial statistical uncertainty at this horizon and underscoring that these projections should be interpreted as probabilistic rather than precise predictions.

Total Live Births. The Drift model projects a continuation of the estimated annual decline of approximately 356 births per year. The resulting 2030 point forecast is approximately 28,200, with prediction intervals widening substantially over the five-year horizon. This forecast is conditional on the persistence of the average rate of change observed over 1960–2025. Policy interventions — such as extended parental leave schemes or housing subsidies for young families — or structural demographic shifts lie outside the model’s assumptions and could alter the trajectory.

Limitations. Both final models are purely univariate and include no demographic covariates. Chen et al. (2018) show that Singapore’s pro-natalist policies have historically targeted third

and higher-order births, whereas the groups most influential for TFR are 20–29-year-old single women and married childless women aged 30–34; incorporating such subgroup dynamics would require a richer covariate model. A natural extension within the time series framework would be to model TLB as a function of TFR — a conditional-forecast approach demonstrated in the companion analysis — exploiting the Pearson correlation of 0.847 between the two series. The annual data frequency precludes seasonal decomposition, and the 10-year holdout has limited power to discriminate between models with similar accuracy. Structural breaks at the sharp 1980s and 2000s policy shifts are not explicitly modelled.

9 Conclusion

This report examined Singapore’s annual Total Fertility Rate and Total Live Births from 1960 to 2025 using time series methods covered in the MATH X313 curriculum. Both series display persistent long-run decline and strong level autocorrelation ($ACF_1 = 0.91$ and 0.84 respectively), requiring first differencing before ARIMA estimation.

For TFR, **ARIMA(0,1,1)** was selected on the basis of the best combination of holdout forecast accuracy ($RMSE = 0.178$, $MAE = 0.153$) and clean residual diagnostics (Ljung-Box $p = 0.346$). The fitted MA(1) coefficient of -0.799 captures a partial reversion of shocks within one year.

For Total Live Births, the **Drift model** was selected. The linear trend model’s superior raw accuracy was disqualified by a severe violation of the residual independence assumption ($LB = 109.6$, $p < 0.001$), which renders its prediction intervals statistically invalid. The Drift model passes all diagnostic checks, captures the long-run declining trend, and achieves the best holdout RMSE (4,097) among diagnostically adequate candidates.

Five-year forecasts suggest that TFR will continue to decline modestly below 0.87, and total live births will fall toward approximately 28,200 by 2030, with wide uncertainty bands reflecting the inherent difficulty of demographic forecasting. Future work should incorporate TFR as a covariate for live-birth modelling and investigate whether government pro-natalist policy expenditure constitutes a structural break that improves forecast accuracy.

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